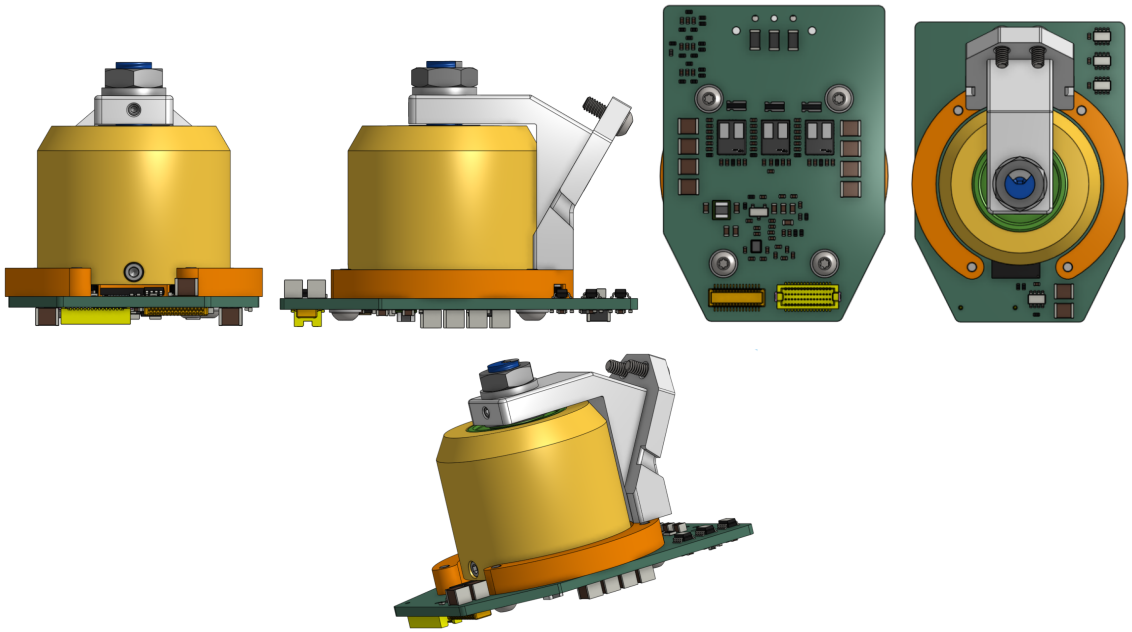
OreSat Reaction Wheel Datasheet

For the v4.1 wheels + v8.1.1 board
2026-11-03



Electrical and Mechanical Requirements

- Voltage: (6.0, 7.2, 8.4) V - meant for 2S Li pack; 12 V abs max
- Current: 2 A (4A peak)
- Power
    - 0.5 W idle
    - 1.3 W @ 5 krpm hold
    - 1.9 W@ 8 krpm hold
    - 6.2 W @ 30 krpm hold
    - 25 W peak
- Weight (wheel + board + brackets): 109 g
- Dimensions: 50 x 36 x 37.1 mm

System Performance

- Total angular momentum: 23 mNms
- Nominal Torque: 0.9 mNm at default 20 rev/s^2 acceleration
- Maximum torque (calculated): 6.5 mNm
- Maximum angular speed: 500 rev/s (30k rpm)

Data Interface

- CANopen interface over 1 Mbps CAN 2.0.
- Firmware updates over CAN.

- UART available, but no current interface.
- SWD to program STM32G474 microcontroller.

Source

- Mechanical CAD
- Electrical CAD
- Firmware

Notes

- Wheels uses a modified ECO Micro Series 1106 6000 kV motor.